

Fast Curvature Baker User Guide

Fast Curvature Baker is a high-speed GPU-accelerated curvature map baking tool for Unity.

Easily generate edge wear, crevice dirt/shadow masks, and smooth curvature gradient maps for 3D meshes with a single click.

UI Overview

Fast Curvature Baker v1.0.0 最新版を取得できません 

EN

対象オブジェクト

選択中を追加

クリア

GameObject をドラッグ&ドロップして追加

ベイク基本設定

ベイクモード

Default

▼

解像度

2048

▼

UVチャンネル

UV0

▼

品質

Standard

▼

スーパーサンプリング 2x

☒

ハードエッジ (エッジ検出)

エッジ幅 (m)

0.003

エッジ強度

1

曲面設定 (曲面検出)

曲面計測半径 (m)

0.01

曲面強度

1

計算ソース

ShadingNormals

▼

▶ 詳細設定

出力設定

同名ファイルを上書き保存する ☐

出力先: <ベーステクスチャのフォルダ>/BakedCurvature/ (テクスチャがない場合は Assets/BakedCurvature/)



Getting Started

Open the window from the Unity top menu: **dennokoworks > Fast Curvature Baker**.

Features & Settings

1. Top Bar (Header)

- **Title:** **Fast Curvature Baker**
- **Version Display:** Shows the current tool version (e.g., **v1.0.0**). It automatically checks for updates on startup.
 - **Update Available <Version>**: Displayed when a newer release is found.
 - **Could not check for updates**: Displayed if a network or retrieval error occurs.
- **Refresh Button (↻)**: Manually retries the version check.
- **Language Button ([EN] / [JA])**: Switches the UI language between English and Japanese with a single click (preference is saved).

2. Target Objects

Specifies the meshes to bake (**GameObjects** containing a **MeshRenderer** or **SkinnedMeshRenderer**).

- **Add Selected**: Adds the currently selected **GameObjects** in the Hierarchy or Scene to the list.
- **Clear**: Clears all registered target objects.
- **Drag & Drop Area**: Drag and drop **GameObjects** directly from the Project or Hierarchy window.
- **Target List**: Lists registered **GameObjects**. Each entry has a ✕ button on the right to remove it individually.

3. Bake Settings

Setting	Description	Effect of Changing
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Setting	Description	Effect of Changing
Bake Mode	Selects the type of curvature map to output. • Default: Signed curvature (flat = 0.5, convex = white, concave = black). • Convex: Convex edges only (highlights, scratches, worn edge masks). • Concave: Concave crevices/valleys only (dirt, ambient shadows, dust masks).	Switches between grayscale masks and full signed curvature maps according to your workflow.
Resolution	Pixel resolution of the output texture (256 to 4096).	▲ Increase: Sharper details and crisper lines, but increases bake time and memory usage. ▼ Decrease: Faster baking and smaller file size, with lower detail.
UV Channel	The mesh UV channel to bake onto (UV0 to UV7).	Typically UV0 is used for main material textures.
Quality	Number of sample points per radius query (Draft: 4 points, Standard: 6 points, High: 8 points, Ultra: 12 points).	▲ Increase: Reduces sample noise and improves accuracy, but takes longer to compute. ▼ Decrease: Useful for fast iteration and parameter preview.
Supersample 2x	Computes curvature internally at 2x resolution and downsamples with box averaging (automatically disabled at 4096).	ON: Greatly reduces aliasing (jagged edges) along UV seams and thin features. OFF: Reduces baking time.

4. Hard Edges (Edge Detection)

Detects sharp creases and hard normal breaks between polygon faces.

Setting	Description	Effect of Changing
Edge Width (m)	World-space width (in meters) around sharp edges to detect curvature.	▲ Increase: Widens the detected edge line for broader edge wear. ▼ Decrease: Produces a razor-sharp, thin edge line suitable for crisp mechanical borders.
Edge Strength	Multiplier for hard edge curvature. A value of 1.0 reaches full brightness at a 90-degree angle.	▲ Increase: Makes even gentle creases stand out boldly with high contrast. ▼ Decrease: Softens edge influence. Setting to 0.0 disables hard edge detection.

5. Smooth Surfaces (Surface Detection)

Detects smooth curvature across gentle surfaces and curved geometry.

Setting	Description	Effect of Changing
Radius (m)	World-space radius (in meters) of the sampling sphere used to evaluate curvature.	▲ Increase: Captures broad geometry undulations and forms smooth, global gradients. ▼ Decrease: Responds sensitively to fine micro-details and tight contours.
Strength	Multiplier for smooth surface curvature. A value of 1.0 reaches full brightness for a sphere whose radius equals Radius .	▲ Increase: Enhances contrast on gentle curvatures. ▼ Decrease: Produces softer, subtler gradients.
Source	The normal source used for curvature calculation. • ShadingNormals: Uses vertex/interpolated normals (smooth shaded gradients). • Geometry: Uses actual polygon face normals (directly measures polygon mesh creases).	ShadingNormals: Delivers clean, smooth gradients even on lower-poly models. Geometry: Ideal for high-poly or hard-surface models where you want the mesh facet angles directly captured.

6. Advanced Settings

Click the foldout header to reveal advanced parameters.

Setting	Description	Effect of Changing
Same Part Only	Restricts sample points to the same connected mesh component as the query texel.	ON: Prevents accidental curvature crosstalk between separate overlapping parts (e.g., clothing resting over skin). OFF: Samples adjacent separate mesh parts together as a unified surface.
Normal Rejection	Dot product threshold rejecting sample points facing away from the texel normal (weight smoothly blends back over +0.25).	▲ Increase: Strictly eliminates interference from backfaces and thin sheet geometry. ▼ Decrease: Ensures tight, steep crevasses are fully detected.
Blur Passes	Number of 3x3 separable Gaussian blur iterations applied post-bake (0 to 16 passes).	▲ Increase: Smooths out polygon stepping artifacts and high-frequency noise (slightly softens edges). ▼ Decrease: Preserves maximum sharpness (0 = no blur).
Dilation (px)	Number of pixels to dilate texture data outward from UV island boundaries.	▲ Increase: Prevents dark seam artifacts at mipmapped distances or seams. ▼ Decrease: Prevents overgrowth into closely packed UV islands.

7. Output Settings

- **Overwrite Existing Files:**
 - **Unchecked (Default):** If a file with the same name exists, automatically appends a space and sequence number (e.g., `Model_Curvature 1.png`, `Model_Curvature 2.png`) to avoid overwriting.
 - **Checked:** Overwrites the existing file directly.
 - **Output Destination:**
 - If a base texture is assigned to the material: `<BaseTextureFolder>/BakedCurvature/`
 - If no base texture is found: `Assets/BakedCurvature/`
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8. Action Buttons & Status Bar

- **Bake Curvature Map:** Starts the baking process. A progress bar will be displayed, and the operation can be cancelled at any time.
 - **Reset to Defaults:** Resets all parameters to recommended default values.
 - **Status Bar:** Located at the bottom of the window; displays current state messages such as `Ready`, `Baking...`, save confirmations, or error alerts.
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Use Cases & Tips

1. Avatar & Costume Edge Wear / Chipping

- Set Bake Mode to `Convex` and `Edge Width` to a small value (`0.002 – 0.005`).
- Assign the baked texture to rim light, roughness, or decal mask inputs in shaders like `litToon` or `Poiyomi` to simulate realistic metal edge abrasion or leather wear.

2. Crevice Dirt & Shadow Masks

- Set Bake Mode to `Concave`.
- Multiply the baked texture into your base color or ambient occlusion slot to shade fabric folds, seams, and crevices where dust naturally collects.

3. Substance Painter / 3D Painting Workflows

- Set Bake Mode to `Default`.
- Export the map as a standard curvature input for smart materials, generators, and procedural masks in external 3D texturing suites.